

Topic 7: PCA

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```
library(MASS)
Sigma = matrix(c(1,0.846,0.805,0.859,0.473,0.398,0.301,0.382,
                 0.846,1,0.881,0.826,0.376,0.326,0.277,0.415,
                 0.805,0.881,1,0.801,0.380,0.319,0.237,0.345,
                 0.859,0.826,0.801,1,0.436,0.329,0.327,0.365,
                 0.473,0.376,0.380,0.436,1,0.762,0.730,0.629,
                 0.398,0.326,0.319,0.329,0.762,1,0.583,0.577,
                 0.301,0.277,0.237,0.327,0.730,0.583,1,0.539,
                 0.382,0.415,0.345,0.365,0.629,0.577,0.539,1),nrow = 8, ncol
= 8, byrow = T)

set.seed(123)
sample_data = mvrnorm(n = 50, mu = c(rep(0,8)), S = Sigma)

p = nrow(Sigma)
EQ = eigen(Sigma)
V_PC = EQ$values
P = EQ$vectors
pca_sample = prcomp(sample_data, scale. = FALSE)
TPV = sum(diag(Sigma)) #Total population Variance
PV_PC = round(V_PC/TPV * 100); #percent variance of principal component
CPV_PC = round(cumsum(PV_PC)); #cumulative percent variance of principal
component

# Step 1: Create the original PCA Loadings table
var_names <- paste0("X", 1:p)
loadings <- round(P, 3)
PC_table <- data.frame(Variable = var_names, loadings)
colnames(PC_table)[2:(p+1)] <- paste0("PC", 1:p)

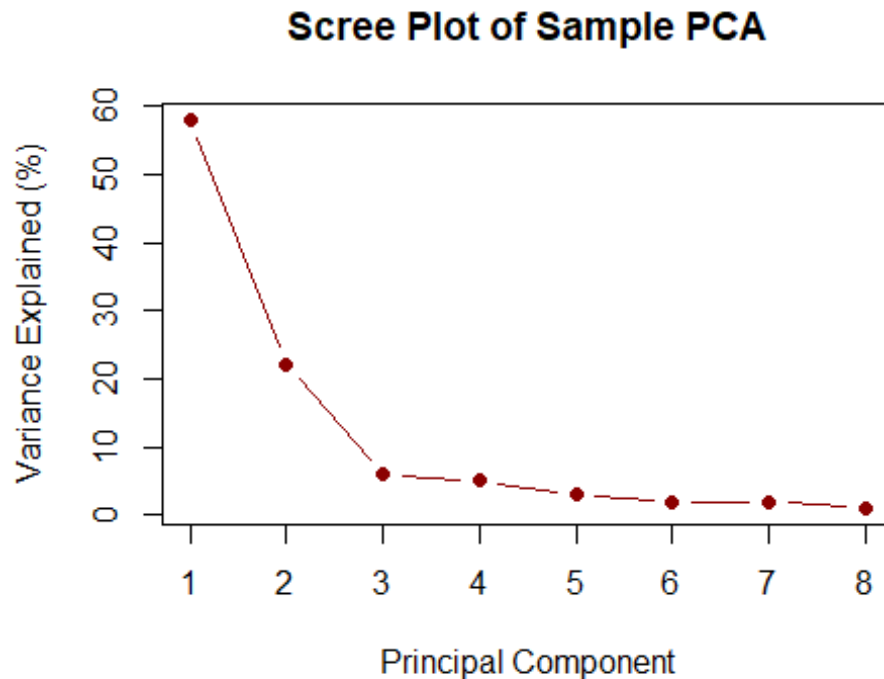
# Step 2: Prepare the summary rows
V_PC_row <- c("V(PC)", round(V_PC, 3))
PV_PC_row <- c("PV(PC)", round(PV_PC, 3))
CPV_PC_row <- c("CPV(PC)", round(CPV_PC, 3))

# Step 3: Combine all into one table
PC_table_updated <- rbind(PC_table, V_PC_row, PV_PC_row, CPV_PC_row)

# Step 4: Display the updated table
print(PC_table_updated)
```

##	Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
## 1	X1	-0.398	-0.28	-0.101	0.107	-0.408	-0.152	0.636	0.384
## 2	X2	-0.389	-0.331	0.113	-0.068	0.341	-0.072	0.278	-0.723
## 3	X3	-0.376	-0.345	0.015	0.047	0.541	0.392	-0.242	0.482

```
# Scree plot
plot(PV_PC, type = "b", pch = 19, col = "darkred",
     xlab = "Principal Component", ylab = "Variance Explained (%)",
     main = "Scree Plot of Sample PCA")
```



PCA on iris dataset.

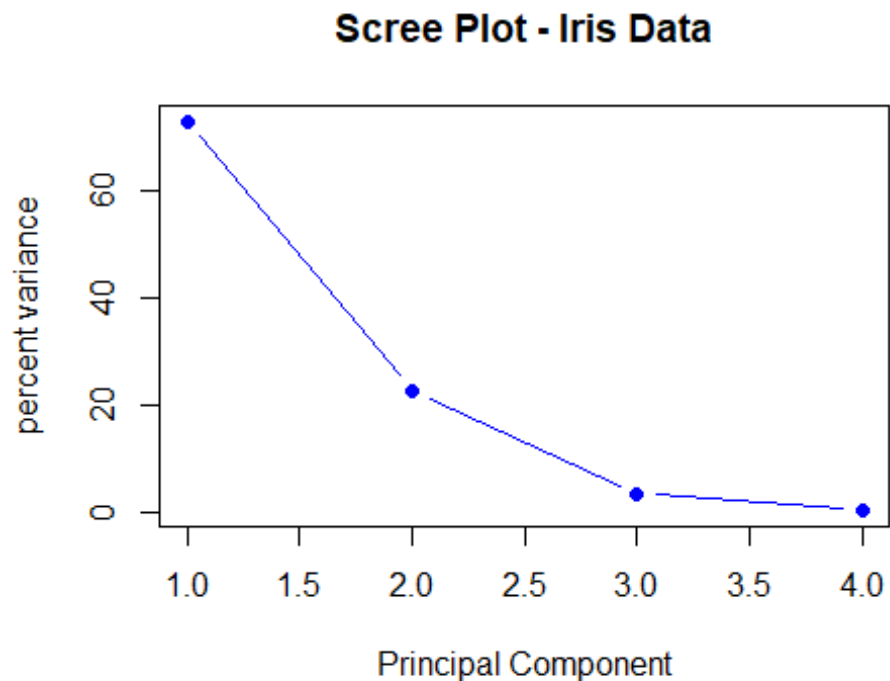
```
# Use iris dataset
data(iris)
iris_num <- scale(iris[, 1:4])

s1 <- cor(iris_num)
ea <- eigen(s1)
v_pc <- round(ea$values, 3)
p <- ea$vectors
TPV <- sum(diag(s1))
pv_pc <- round((v_pc / TPV) * 100, 3)
cpv_pc <- round(cumsum(pv_pc), 3)

pc_summary <- data.frame(
  PC = paste0("PC", 1:length(v_pc)),
  Eigenvalue = v_pc,
  Proportion = pv_pc,
  Cumulative = cpv_pc
)
```

```
pc_df <- cbind(pc_summary, round(p, 4))
print(pc_df)

# ---- Scree plot ----
plot(pv_pc,
     type = "b",
     pch = 19,
     col = "blue",
     xlab = "Principal Component",
     ylab = "percent variance",
     main = "Scree Plot - Iris Data")
```



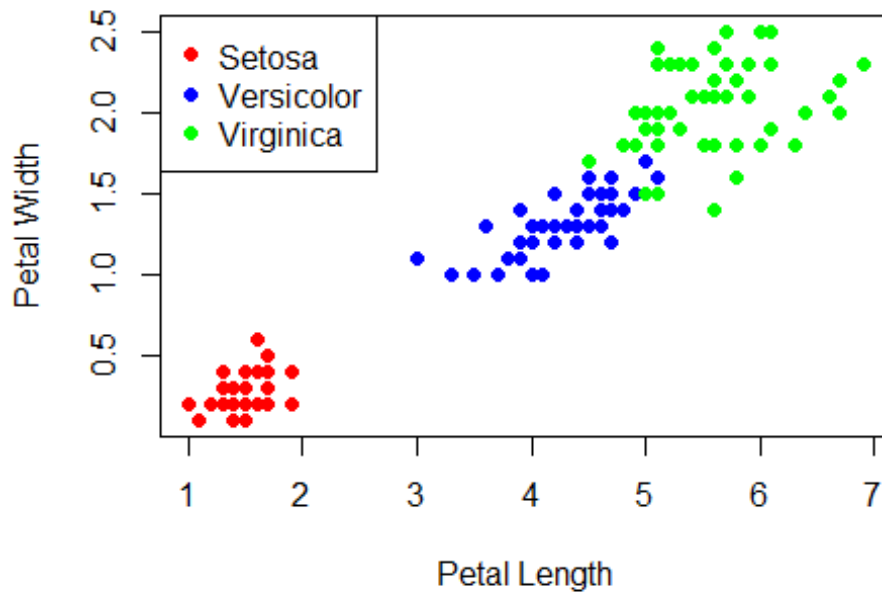
```
data("iris")
iris_subset <- iris[, c("Petal.Length", "Petal.Width", "Species")]
setosa <- subset(iris_subset, Species == "setosa")
versicolor <- subset(iris_subset, Species == "versicolor")
virginica <- subset(iris_subset, Species == "virginica")
plot(
  iris_subset$Petal.Length,
  iris_subset$Petal.Width,
  col = c(rep("red", 50), rep("blue", 50), rep("green", 50)),
  pch = 19,
  xlab = "Petal Length",
  ylab = "Petal Width",
  main = "Iris Petal Length vs Width (by Species Color)"
)
legend("topleft",
```

```

legend = c("Setosa", "Versicolor", "Virginica"),
col = c("red", "blue", "green"),
pch = 19)

```

Iris Petal Length vs Width (by Species Color)



OR code

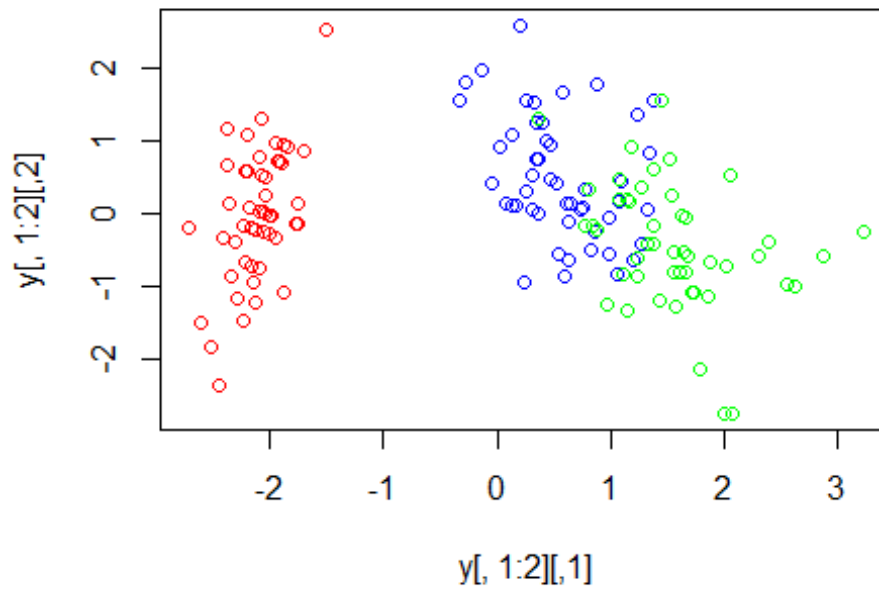
```

#or code for above
library(MASS)
library(Matrix)
x=scale(iris[,1:4])
x

## Sepal.Length Sepal.Width Petal.Length Petal.Width
##      5.843333      3.057333      3.758000      1.199333
## attr(,"scaled:scale")
## Sepal.Length Sepal.Width Petal.Length Petal.Width
##      0.8280661      0.4358663      1.7652982      0.7622377

y=(p)%*%t(x)
y=t(y)
plot(y[,1:2],col=c(rep("red",50),rep("blue",50),rep("green",50)))

```



2.

```
library(expm);library(Matrix);library(matlib)
```

```
S=matrix(c(8,2,3,1,
           2,5,-1,3,
           3,-1,6,-2,
           1,3,-2,7),nrow=4)

p=nrow(S);q=2
n=ncol(S)
#-----manipulation-----
S11=S[1:q,1:q]
S12=S[1:q,(q+1):p]
S21=t(S12)
S22=S[(q+1):p,(q+1):p]
S22_inv=solve(S22)
S11_inv=solve(S11)
A=inv(sqrtm(S22))%%S21%%inv(sqrtm(S11))
R2=A%%t(A)
R1=t(A)%%A

E=eigen(R1)$vectors
L1=eigen(R1)$values

F_=eigen(R2)$vectors
M1=eigen(R2)$values
cbind(L1,M1)
```

```

##           L1           M1
## [1,] 0.153750229 0.153750229
## [2,] 0.003323215 0.003323215

U=inv(sqrtm(S11))%%E
V=inv(sqrtm(S22))%%F_

rho=sqrt(L1)

#-----Output-----
sprintf("rho(1)= First canonical correlation coeff.=%f",rho[1])
## [1] "rho(1)= First canonical correlation coeff.=0.392110"

sprintf("rho(2)= Second canonical correlation coeff.=%f",rho[2])
## [1] "rho(2)= Second canonical correlation coeff.=0.057647"

sprintf("The coefficients of first pair of canonical variables are (in
column)")
## [1] "The coefficients of first pair of canonical variables are (in
column)"

print(cbind(U[,1],V[,1]))

##           [,1]           [,2]
## [1,] -0.06907785 0.06167261
## [2,] 0.46709523 0.39166283

sprintf("The coefficients of second pair of canonical variables are (in
column)")
## [1] "The coefficients of second pair of canonical variables are (in
column)"

print(cbind(U[,2],V[,2]))

##           [,1]           [,2]
## [1,] -0.36622007 -0.42474347
## [2,] 0.06359462 -0.06704449

#-----Verification-----
sprintf("The VCM of canonical variables is:")
## [1] "The VCM of canonical variables is:"

ccmat=t(U)%%S12%%V
print(ccmat)

##           [,1]           [,2]
## [1,] 0.03230926 -0.382391705
## [2,] 0.02544276 -0.003449869

```

```

#-----scatter plot of first pair of canonical variables-----
l=as.matrix(U[,1])
m=as.matrix(V[,1])

# Load required Libraries
library(expm)
library(Matrix)
library(matlib)

# Load iris data
data(iris)

# Split into two variable sets
X <- as.matrix(iris[, 1:2]) # Sepal.Length, Sepal.Width
Y <- as.matrix(iris[, 3:4]) # Petal.Length, Petal.Width

# Compute covariance matrix of combined data
Z <- cbind(X, Y)
S <- cov(Z)

# Dimensions
p <- nrow(S)
q <- ncol(X)
n <- ncol(S)

# Partition covariance matrix
S11 <- S[1:q, 1:q]
S12 <- S[1:q, (q+1):p]
S21 <- t(S12)
S22 <- S[(q+1):p, (q+1):p]

# Matrix manipulation
A <- inv(sqrtm(S22)) %*% S21 %*% inv(sqrtm(S11))
R1 <- t(A) %*% A
R2 <- A %*% t(A)

# Eigen decomposition
E <- eigen(R1)$vectors
L1 <- eigen(R1)$values

F_ <- eigen(R2)$vectors
M1 <- eigen(R2)$values

# Canonical coefficients
U <- inv(sqrtm(S11)) %*% E
V <- inv(sqrtm(S22)) %*% F_

# Canonical correlations
rho <- sqrt(L1)

```

```

# Output
sprintf("rho(1)= First canonical correlation coeff.=%f", rho[1])
## [1] "rho(1)= First canonical correlation coeff.=0.940969"

sprintf("rho(2)= Second canonical correlation coeff.=%f", rho[2])
## [1] "rho(2)= Second canonical correlation coeff.=0.123937"

sprintf("The coefficients of first pair of canonical variables are (in
column):")

print(cbind(U[,1], V[,1]))

##           [,1]      [,2]
## [1,] -1.0689812 -0.8491249
## [2,]  0.8549914  0.6938022

sprintf("The coefficients of second pair of canonical variables are (in
column):")

## [1] "The coefficients of second pair of canonical variables are (in
column):"

print(cbind(U[,2], V[,2]))

##           [,1]      [,2]
## [1,] -0.5797395  1.918708
## [2,] -2.1462750 -4.809531

# Verification: cross-covariance of canonical variables
sprintf("The VCM of canonical variables is:")

## [1] "The VCM of canonical variables is:"

ccmat <- t(U) %*% S12 %*% V
print(ccmat)

##           [,1]      [,2]
## [1,]  9.409690e-01 -8.881784e-16
## [2,] -3.469447e-18  1.239369e-01

# Scatter plot of first canonical pair
l <- X %*% U[,1]
m <- Y %*% V[,1]
plot(l, m, main = "First Canonical Correlation", xlab = "U1", ylab = "V1",
col = iris$Species)

```

First Canonical Correlation

